Diophantine Equations, SFFT, Bezout & Floor Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Diophantine Equations, SFFT, Bezout & Floor Functions Practice Exercises

1 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 1x + 2y = 10$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 6

b. 5

c. 3 ✓

d. 2

Add $1 \cdot 2 = 2$ to both sides (SFFT): $(x + 2)(y + 1) = 10 + 2 = 12$. For $x, y \geq 1$, we need $d = x + 2 \geq 3$ and $12/d = y + 1 \geq 2$. Counting divisors $d \mid 12$ satisfying both inequalities yields exactly 3 positive integer pairs.

2 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 2x + 3y = 18$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 8

b. 3 ✓

c. 5

d. 2

Add $2 \cdot 3 = 6$ to both sides (SFFT): $(x + 3)(y + 2) = 18 + 6 = 24$. For $x, y \geq 1$, we need $d = x + 3 \geq 4$ and $24/d = y + 2 \geq 3$. Counting divisors $d \mid 24$ satisfying both inequalities yields exactly 3 positive integer pairs.

3 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 1x + 3y = 17$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 3 ✓

b. 6

c. 5

d. 2

Add $1 \cdot 3 = 3$ to both sides (SFFT): $(x + 3)(y + 1) = 17 + 3 = 20$. For $x, y \geq 1$, we need $d = x + 3 \geq 4$ and $20/d = y + 1 \geq 2$. Counting divisors $d \mid 20$ satisfying both inequalities yields exactly 3 positive integer pairs.

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4 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 2x + 5y = 26$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 9

b. 5

c. 2

d. 3 ✓

Add $2 \cdot 5 = 10$ to both sides (SFFT): $(x + 5)(y + 2) = 26 + 10 = 36$. For $x, y \geq 1$, we need $d = x + 5 \geq 6$ and $36/d = y + 2 \geq 3$. Counting divisors $d \mid 36$ satisfying both inequalities yields exactly 3 positive integer pairs.

5 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$?

a. 4

b. 5

c. 9 ✓

d. 8

Clearing denominators gives $6(x + y) = xy \Leftrightarrow xy - 6x - 6y = 0$. Adding $6^2 = 36$ via SFFT gives $(x - 6)(y - 6) = 36$. Since $x, y > 6$ for all positive solutions, $x - 6$ ranges bijectively over all positive divisors of $6^2 = 36$, giving $\tau(36) = 9$ ordered pairs.

6 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{8}$?

a. 4

b. 7 ✓

c. 6

d. 8

Clearing denominators gives $8(x + y) = xy \Leftrightarrow xy - 8x - 8y = 0$. Adding $8^2 = 64$ via SFFT gives $(x - 8)(y - 8) = 64$. Since $x, y > 8$ for all positive solutions, $x - 8$ ranges bijectively over all positive divisors of $8^2 = 64$, giving $\tau(64) = 7$ ordered pairs.

7 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{9}$?

a. 5 ✓

b. 3

c. 4

d. 6

Clearing denominators gives $9(x + y) = xy \Leftrightarrow xy - 9x - 9y = 0$. Adding $9^2 = 81$ via SFFT gives $(x - 9)(y - 9) = 81$. Since $x, y > 9$ for all positive solutions, $x - 9$ ranges bijectively over all positive divisors of $9^2 = 81$, giving $\tau(81) = 5$ ordered pairs.

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8 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{10}$?

a. 4

b. 5

c. 8

d. 9 ✓

Clearing denominators gives $10(x + y) = xy \Leftrightarrow xy - 10x - 10y = 0$. Adding $10^2 = 100$ via SFFT gives $(x - 10)(y - 10) = 100$. Since $x, y > 10$ for all positive solutions, $x - 10$ ranges bijectively over all positive divisors of $10^2 = 100$, giving $\tau(100) = 9$ ordered pairs.

9 Find the LARGEST positive integer n such that $n + 2$ divides $n^2 + 3n + 15$.

a. 13

b. 4

c. 11 ✓

d. 12

Perform polynomial division by $n + 2$: $n^2 + 3n + 15 = (n + 2)(n + 1) + (2^2 - 3(2) + 15) = (n + 2)(n + 1) + 13$. Thus $(n + 2) \mid (n^2 + 3n + 15) \Leftrightarrow (n + 2) \mid 13$. The largest divisor of 13 is 13 itself, giving $n + 2 = 13 \Rightarrow n = 11$.

10 Find the LARGEST positive integer n such that $n + 3$ divides $n^2 + 3n + 17$.

a. 17

b. 14 ✓

c. 5

d. 15

Perform polynomial division by $n + 3$: $n^2 + 3n + 17 = (n + 3)(n + 0) + (3^2 - 3(3) + 17) = (n + 3)(n + 0) + 17$. Thus $(n + 3) \mid (n^2 + 3n + 17) \Leftrightarrow (n + 3) \mid 17$. The largest divisor of 17 is 17 itself, giving $n + 3 = 17 \Rightarrow n = 14$.

11 Find the LARGEST positive integer n such that $n + 4$ divides $n^2 + 3n + 19$.

a. 19 ✓

b. 23

c. 7

d. 20

Perform polynomial division by $n + 4$: $n^2 + 3n + 19 = (n + 4)(n + -1) + (4^2 - 3(4) + 19) = (n + 4)(n - 1) + 23$. Thus $(n + 4) \mid (n^2 + 3n + 19) \Leftrightarrow (n + 4) \mid 23$. The largest divisor of 23 is 23 itself, giving $n + 4 = 23 \Rightarrow n = 19$.

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12 Find the LARGEST positive integer n such that $n+5$ divides $n^2+3n+21$.

a. 31

b. 10

c. 27

d. 26 ✓

Perform polynomial division by $n+5$: $n^2+3n+21=(n+5)(n+-2)+(5^2-3(5)+21)=(n+5)(n-2)+31$. Thus $(n+5)\mid(n^2+3n+21)\Leftrightarrow(n+5)\mid 31$. The largest divisor of 31 is 31 itself, giving $n+5=31\Rightarrow n=26$.

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